Embryo Image Preprocessing Pipeline Guide

Optical Sharpness Filtering, Blastocyst Centering, CLAHE Enhancement & Web Integration

1. Why Preprocessing is Essential in Embryo Microscopy

Microscopic blastocyst imaging presents unique challenges that degrade raw neural network accuracy if not preprocessed:

- **Multi-Focal Z-Stack Blur:** Optical lenses photograph embryos at multiple focal depths; off-axis slices are blurry.
- **Peripheral Dark Vignetting:** Inverted microscopes suffer from peripheral falloff and dark borders that mislead CNN filters.
- **Micro-Cellular Low Contrast:** Blastomere membranes and Inner Cell Mass borders require contrast expansion to be discernible.

2. The 4-Step Preprocessing Pipeline

Stage	Algorithm / Method	Purpose & Clinical Benefit
Step 1: Auto-Focus Sharpness	Variance of Laplacian: cv2.Laplacian(img, CV_64F).var()	Calculates optical focus score. When a patient has multiple Z-slices, automatically selects the sharpest equatorial slice.
Step 2: Blastocyst Centering	Aspect-Preserved Square Crop: sz = min(h, w) cropped = img[y:y+sz, x:x+sz]	Centers the embryo circle and eliminates peripheral camera vignetting and dark petri dish borders without squishing aspect ratios.
Step 3: High-Order Resizing	Lanczos-4 Interpolation to 256x256	Provides anti-aliased spatial downsampling, preserving delicate trophectoderm cellular membranes.
Step 4: CLAHE Contrast	LAB Color Space CLAHE: clipLimit=2.0, tileGrid=(8,8) applied to L-channel.	Locally amplifies cellular boundary contrast so the Inner Cell Mass (ICM) and blastocoel junction pop out distinctly.

3. How to Use in Website Code (Python)

The companion script preprocessing_pipeline.py is ready to import into your website backend:

```
from preprocessing_pipeline import preprocess_embryo_image, compute_sharpness

# 1. Preprocess any uploaded embryo image (file path, PIL, or numpy array):
clean_image = preprocess_embryo_image('user_upload.png', target_size=256)

# 2. If the user uploads multiple slices of an embryo (Z-stack):
from preprocessing_pipeline import select_best_slice
sharpest_slice_path = select_best_slice(['slice1.png', 'slice2.png', 'slice3.png'])
```

4. Backend Deliverables Included in this Subfolder

- preprocessing_pipeline.py: Standalone library with sharpness calculation, square cropping, and CLAHE.
- inference_pipeline.py: Full end-to-end predictor that loads the 4 models and returns ready-to-display JSON for your web app.
- models/: Directory containing all 4 trained model checkpoints.